

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Assessment Tool Weightage: 10%

Compute the required performance measures from the given data and compare the techniques based on the results. Evaluate and justify your conclusion using the calculated values.

1. Detection Performance Comparison

Two object detection models produce the following results:

Model	True Positives	False Positives	False Negatives
A (Viola-Jones)	80	20	30
B (YOLO)	90	15	20

(a) Compute precision and recall for both models.

(b) Compare their performance and state which model is better for object detection.

ANSWER:

(a) Performance Measures

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}$$

- Model A (Viola-Jones):
 - $\text{Precision}_A = \frac{80}{80+20} = \frac{80}{100} = 80.00\%$
 - $\text{Recall}_A = \frac{80}{80+30} = \frac{80}{110} \approx 72.73\%$
- Model B (YOLO):
 - $\text{Precision}_B = \frac{90}{90+15} = \frac{90}{105} \approx 85.71\%$

$$\circ \text{ Recall}_B = \frac{90}{90+20} = \frac{90}{110} \approx 81.82\%$$

Metric	Model A (Viola-Jones)	Model B (YOLO)	Advantage
Precision	80.00%	85.71%	Model B (+5.71%)
Recall	72.73%	81.82%	Model B (+9.09%)

{b) Evaluation & Conclusion

Model B (YOLO) is superior. It achieves both higher precision (fewer false positives) and higher recall (fewer missed detections), making it significantly more reliable overall.

2. Motion Estimation Error Analysis

Two motion estimation techniques produce displacement errors (in pixels):

Frame Method A Method B

1	2	3
2	3	5
3	2	4

(a) Compute average error for both methods.

(b) Compare and evaluate which method is more accurate.

ANSWER:

(a) Average Error Computation

$$\text{Average Error} = \frac{\sum \text{Displacement Error}}{N}$$

- Method A: Average Error_A = $\frac{2+3+2}{3} = \frac{7}{3} \approx 2.33$ pixels
- Method B: Average Error_B = $\frac{3+5+4}{3} = \frac{12}{3} = 4.00$ pixels

(b) Evaluation & Conclusion

Method A is more accurate. It reduces mean displacement error by 1.67 pixels (41.8% lower error rate compared to Method B).

4. Depth Estimation Comparison (Stereo vs SfM)

Estimated depth values (in meters):

Object Stereo Vision Structure from Motion

A	2.0	2.5
B	3.0	3.8
C	4.0	5.0

Actual depths: A=2, B=3, C=4

(a) Compute absolute error for both methods.

(b) Compare and evaluate which method is more accurate.

ANSWER:

(a) Absolute Error Computation

$$\text{Absolute Error} = |\text{Estimated Depth} - \text{Actual Depth}|$$

- **Stereo Vision:**
 - **Object A:** $|2.0 - 2.0| = 0.0 \text{ m}$
 - **Object B:** $|3.0 - 3.0| = 0.0 \text{ m}$
 - **Object C:** $|4.0 - 4.0| = 0.0 \text{ m}$
 - **Mean Absolute Error (MAE):** 0.00 m
- **Structure from Motion (SfM):**
 - **Object A:** $|2.5 - 2.0| = 0.5 \text{ m}$
 - **Object B:** $|3.8 - 3.0| = 0.8 \text{ m}$
 - **Object C:** $|5.0 - 4.0| = 1.0 \text{ m}$
 - **Mean Absolute Error (MAE):** $\frac{0.5+0.8+1.0}{3} \approx 0.77 \text{ m}$

Object	Actual Depth (m)	Stereo Estimated (m)	Stereo Error (m)	SfM Estimated (m)	SfM Error (m)
A	2.0	2.0	0.0	2.5	0.5
B	3.0	3.0	0.0	3.8	0.8
C	4.0	4.0	0.0	5.0	1.0
MAE	—	—	0.00	—	0.77

(b) Evaluation & Conclusion

Stereo Vision is accurate (zero error across all test points), whereas SfM exhibits a systematic positive bias that increases with object distance.

8. Optical Flow Consistency

Motion values:

Frame	Method A	Method B
1	4	6
2	5	8
3	6	9

- (a) Compute average motion.
- (b) Compare consistency and evaluate which is more stable.

ANSWER:

(a) Average Motion Computation

$$\mu = \frac{\sum \text{Motion Values}}{N}$$

- Method A: $\mu_A = \frac{4+5+6}{3} = 5.00 \text{ units}$
- Method B: $\mu_B = \frac{6+8+9}{3} \approx 7.67 \text{ units}$

(b) Consistency & Stability Evaluation

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

- **Method A Variance:** $\sigma_A^2 = \frac{(4-5)^2 + (5-5)^2 + (6-5)^2}{3} = \frac{1+0+1}{3} \approx 0.67$
- **Method B Variance:** $\sigma_B^2 = \frac{(6-7.67)^2 + (8-7.67)^2 + (9-7.67)^2}{3} \approx \frac{2.78+0.11+1.78}{3} \approx 1.56$

Method A is significantly more consistent and stable, demonstrating a variance less than half that of Method B (0.67 vs. 1.56).

10. Error Rate Comparison

Results:

Method	Errors	Total
A	20	100
B	10	100

- (a) Compute error rate.
- (b) Compare and evaluate.

ANSWER:

(a) Error Rate Computation

$$\text{Error Rate} = \frac{\text{Errors}}{\text{Total Samples}} \times 100\%$$

- Method A: $\frac{20}{100} = 20.00\%$
- Method B: $\frac{10}{100} = 10.00\%$

(b) Evaluation & Conclusion

Method B is the superior approach, cutting the error rate in half (10% vs. 20%) and doubling operational reliability under identical sample sizes.

11. Motion Estimation Variance

Errors:

Frame	A	B
1	2	6
2	3	7
3	2	5

- (a) Compute mean error.
- (b) Compare variability and evaluate stability.

ANSWER:

(a) Mean Error Computation

- **Method A:** $\mu_A = \frac{2+3+2}{3} \approx 2.33$ pixels

- **Method B:** $\mu_B = \frac{6+7+5}{3} = 6.00$ pixels

(b) Variability & Stability Evaluation

$$\sigma^2 = \frac{\sum(e_i - \mu)^2}{N}$$

- **Method A Variance:** $\sigma_A^2 = \frac{(2-2.33)^2 + (3-2.33)^2 + (2-2.33)^2}{3} \approx 0.22 \text{ pixels}^2$
- **Method B Variance:** $\sigma_B^2 = \frac{(6-6.0)^2 + (7-6.0)^2 + (5-6.0)^2}{3} \approx 0.67 \text{ pixels}^2$

Parameter	Method A	Method B	Optimal Target
Mean Error	2.33 pixels	6.00 pixels	Lower
Variance (σ^2)	0.22 pixels ²	0.67 pixels ²	Lower

Method A is both more accurate (lower mean error) and more stable (three times lower variance across frames).

14. Depth Estimation Error (Squared)

Predicted:

Object	A	B	Actual
1	2.5	2.2	2
2	3.5	3.1	3
3	4.5	4.2	4

- Compute squared error.
- Compare and evaluate accuracy.

ANSWER:

(a) Squared Error Computation

$$SE = (\text{Predicted} - \text{Actual})^2, \quad MSE = \frac{\sum SE}{N}$$

- Object 1 (Actual = 2.0):
 - Model A: $(2.5 - 2.0)^2 = 0.25 \text{ m}^2$
 - Model B: $(2.2 - 2.0)^2 = 0.04 \text{ m}^2$
- Object 2 (Actual = 3.0):
 - Model A: $(3.5 - 3.0)^2 = 0.25 \text{ m}^2$
 - Model B: $(3.1 - 3.0)^2 = 0.01 \text{ m}^2$
- Object 3 (Actual = 4.0):
 - Model A: $(4.5 - 4.0)^2 = 0.25 \text{ m}^2$
 - Model B: $(4.2 - 4.0)^2 = 0.04 \text{ m}^2$
- Mean Squared Error (MSE):
 - $MSE_A = \frac{0.25+0.25+0.25}{3} = 0.250 \text{ m}^2$
 - $MSE_B = \frac{0.04+0.01+0.04}{3} = 0.030 \text{ m}^2$

(b) Evaluation & Conclusion

15. Recognition Precision Comparison

Results:

System	TP	FP
A	90	10
B	85	5

(a) Compute precision.

(b) Compare and evaluate performance.

ANSWER:

(a) Precision Computation

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

- **System A:** $\text{Precision}_A = \frac{90}{90+10} = \frac{90}{100} = 90.00\%$
- **System B:** $\text{Precision}_B = \frac{85}{85+5} = \frac{85}{90} \approx 94.44\%$

(b) Evaluation & Conclusion

System B outperforms **System A** (94.44% vs. 90.00%). By reducing false positives by 50% (5 vs. 10), **System B** provides higher prediction confidence.

Code of Conduct:

I DARANI E (192312120) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation ; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand